



METTU UNIVERSITY BEDELE CAMPUS

**DETERMINANT OF TEFF COMMERCIALIZATION IN THE CASE OF
BEDELE DISTRICT BUNO BEDELE ZONE OROMIA ETHIOPIA**

**SENIOR RESEARCH PROPOSAL SUBMITTED TO METTU
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BUSINESS AND VALUE CHAIN MANAGEMENT**

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ABSTRACT

The main purpose of this study was to evaluate the determinants of teff commercialization in Bedele district. It is widely accepted that teff commercialization is one of the objective of farmers to fulfill their family basic needs. Having recognized this, the proposal has conducted this study by focusing on the teff commercialization and the influencing factors. To this end, data were collected from primary sources. 94 individuals were randomly selected in proportion to the membership size of the teff commercialization. The data were collected through questionnaire. Both descriptive statistics and econometric analysis were employed for analytical purpose. The result shows that from the total respondents, 46.8 % were female who commercialize teff while 53.2%, were male teff commercializes. In the regression analysis, variables like age of house hold , access to credit ,intensity of fertilizer and farm experiences were found to have significant relationship to the commercialization of teff. Among these significant variables access to credit, intensity of fertilizer ,and farm experience have positive relationship while age of house hold have negative relationship with teff commercialization. Based on the result, responsible producers need to give due attention to overcome strengthen, build their capacities mainly in teff commercializing and decision making that ultimately could increase their production of teff.

ABBREVIATION AND ACRONOMY

CSA	Central Statistics Agency
GDP	Gross Domestic Product
GVCP	Gross Value of all Crop Production
GVCS	Gross Value of Crop Sales
MoARD	the Ministry of Agriculture and Rural Development
PASDEP	Plan for Accelerated and Sustainable Development to End Poverty
PRSP	Poverty Reduction Strategy Paper
RMA	Rapid Market Appraisal

1. INTRODUCTION

1.1 Background

Agriculture continues to dominate the national economy of Ethiopia, accounting for 36.7% of overall GDP and 70% of foreign exchange earnings. The sector provides employment for 72.7% of the population and is a means of generating livelihood for about 83% of the rural population (ATA, 2017; ADEA, 2014; FAO, 2015). In Ethiopia 95% of the total area under agriculture is cultivated by smallholder farmers and contributes to 90% of the total agricultural output indicating the dominant contribution of smallholder farmers to the overall agricultural production (MoARD, 2010; Gebreslassie and Bekele, 2012). According to MoFED (2010, 2015), the Ethiopian government, in its two-consecutive five-years Growth and Transformation Plan (GTP-I and GTP-II), has given much emphasis on agricultural commercialization, among which the second pillar intends to achieve growth and thereby improve people's livelihoods and reduce poverty.

Commercialization of the smallholder farmers has been viewed by the government as the major source of agricultural growth in Ethiopia. The government of Ethiopia implemented agricultural commercialization clusters with the primary goal of commercialization of smallholders' agriculture and agro-industrial development, offering a strategic entry point for private sector engagement (Pauw, 2017). Commercialization entails agricultural production decision intended for market based on market signals and produce offered for sale and use of purchased inputs (Berhanu and Moti, 2010). The welfare gain of agricultural commercialization can be achieved through specialization, comparative advantage, economies of scale and flow of ideas due to regular interaction (Barrett, 2008). In the agricultural sector, cereals cover about 80% of the total grain crop area (9.97 million hectares) and contribute about 87% (23.1 million tons) of the grain production (CSA, 2016). Among cereals, teff (*Eragrostis teff*) stands first in terms of land area, followed by maize and wheat (CSA, 2016). Ethiopia is the center of both origin and diversity for teff (Vavilov, 1951). Teff is a staple food and one of the most important crops for generating farm income, cultural heritage, national identity and nutritional security. The study area is found in Buno Bedele zone of Oromia region, south western Ethiopia. There is a dearth of information in terms of identifying the determinants of smallholders' commercialization of teff producer

particularly in Bedele district Buno Bedele zone of Oromia region, one of the potential areas of teff production in south western Ethiopia. Such information is essential for making knowledge-based decision that are geared towards improving market participation of farmers in teff and contribute to the national development goals of eradicating poverty and improving food security. However, the supply of teff in the study area still can't satisfy the existing market demand and the farmers are not benefited from teff price increment. Since teff is the most economically and socially crucial crop, there is a strong need to address the prevailing information gap and contribute to proper understanding of determinants of commercialization of smallholder farmers in Bedele district. Such information is also required to contribute to the success of GTP-II plan of the country through improved decision of smallholder farmers in teff production and marketing. Therefore, this study analyzes the level of commercialization and identify factors affecting the level of commercialization of teff producers.

1.2. Statement of the Problem

In Ethiopia, small-scale subsistence farmers dependent on low input, rain-fed mixed farming agriculture dominated with traditional technologies accounts for about 95% of the output (CSA,2016). Agricultural production and productivity is very low and the growth in agricultural output has teff kept pace with human population growth. This small-scale subsistence agriculture remains by far the most important sector in Ethiopian economy and directly supports about 85% of the population in terms of employment and livelihood; contributes about 50% of the country's gross domestic product (GDP); and supplies around 73% of raw material requirement of agro-based domestic industries. It is also the major source of food for the population and hence the prime contributing sector to food security. In addition, agriculture is expected to play a key role in generating surplus capital to speed up the country's overall socio-economic development, (Gebreslassie,2012). Agricultural marketing is a very important factor in economic development and lack of a well-functioning agricultural market and marketing system severely hinders the increase of social welfare, income distribution, and food security of developing countries. Moreover markets and marketing system do not develop simultaneously with economic growth. Markets and marketing system should be organized deliberately to enable economic development (Leykun,2014).

Improved information and marketing facility enables farmers to plan their production more in line with market demand, to schedule their harvest at the most profitable time, to decide which market to sell their produce to and negotiate on a more even footing with traders and it also enables traders to move their produce profitably from a surplus to deficit market and to make decisions about the economics of storage, where technically possible. Thus the market information is critical to the law of one price and to the price discovery process (Pauw,2017).

Provision of improved and high yielding varieties, chemical fertilizer, pesticides and insecticides may favor the farmer in increasing production; however, this is not an end by itself. Therefore, Osmani and Hossain, (2014) without modern marketing system, including communications, transportation, storage facilities and financial arrangement this is not possible.

The efforts of increasing agricultural production and productivity have to be accompanied by a well-performing marketing system which satisfies consumer demands with the minimum margin between producers and consumer prices. Higher prices for producer can encourage farmers to adopt new technologies, increase production, (MoFED,2015). However, there are external and internal problems that influence the marketing efficiency in Ethiopia. This has to do with lack of pertinent market information, development of marketing institutions and marketing infrastructure such as storage, transportation etc.

The possible increment in output resulting from the introduction of improved technology could not be exploited in the absence of well-functioning marketing system. An efficient, integrated, and responsive market mechanism is of critical importance for optimal allocation of resources in agriculture and in stimulating farmers to increase their output (Mebrahtom,2014). A well-functioning marketing system is not limited to stimulation but it also increases production by seeking additional output.

In Ethiopia, agricultural growth induces higher overall growth than non-agricultural sectors. This leads to faster poverty reduction since it generates proportionately more income for farm households who represent the bulk of the poor. From within agriculture, staple crops have stronger growth linkage resulting from more than proportionate increase in total GDP. Moreover, such growth linkage becomes stronger overtime (Diao et al, 2007). Supply of agricultural crop in the study area is subjected to seasonal variation where surplus supply at harvest is the main

feature. The nature of the product on the one hand and lack of properly functioning marketing system on the other, often resulted in lower producers' price. , barely, teff, wheat and bean are the major cash crops grown in the study area mainly for market. However, marketing aspects of only barely and bean were undertaken by Osmani and Hossain (2014) while leaving marketing of teff which have potential production volume and marketability, unresolved. Moreover there is a need to employ a market chain approach to fully understand and resolve the problem of teff and wheat at all levels. Yet there is no such study which tries to look into the whole spectrum of marketing chain of these crops and determinants of their supply Bedele Woreda. This makes the undertaking commercialization of teff and wheat in the Woreda imperative. This study is designed to address the prevailing information gap on the subject and contribute to proper understanding of the challenges and assist in developing improved market development strategies to benefit of smallholder farmers, traders, and other market participants.

1.3. Objective of the study

1.3.1. General objective

- ✓ To analyse determinants of commercialization of teff in the study area.

1.3.2 Specific objectives

- ✓ To estimate teff commercialization small holder in the study area.
- ✓ To analysis the major determinants of teff commercialization in the study area.

1.4 Research Questions

This study was attempted to answer the following research questions:

- ✓ What is the level of teff commercialization in Bedele district?
- ✓ What are the determinants of teff commercialization in Bedele district?

1.5. Scope of the Study

Evaluating the role of in grain marketing, economic and social contributions of farmers at rural level is main task of this research hypothesis. The analysis of teff commercialization role at regional and national level critically depends on response parameters from individual farmer

members and societies. Thus, the purpose of this study is to contribute to the analysis and study of the teff commercialization in Bedele district. Moreover, it will help as an input for researchers who want to engage in further study and address the needs and problems of the farmers.

1.6. Significance of the Study

This study focuses on the determinants of teff commercialization and constraints of teff production in Bedele district. The information is expected to assist the teff producer to understand the production potential, and analyze the performance of teff commercialization activities which could serve as a major input to formulate appropriate marketing policies and strategies in Bedele district by identifying interventions that improve efficiency of the marketing system. The study can also serve as an additional source to conduct detailed studies by identifying research agenda.

The study was generate valuable information on teff commercialization that might assist policy makers at various levels to make relevant decisions to intervene in the development of teff commercialization and designing of appropriate policies and strategies. The findings of the study might also be useful to government and non-governmental organization input suppliers, producers, traders, consumers, and marketing agents to make their respective decisions. It may also serve as a reference material for further research on similar topics and other related subjects.

2. LITERATURE REVIEW

2.1 Theoretical review

2.1.2 Historical background of commercialization

The poverty-reduction strategy adopted by Ethiopia seeks to achieve growth through the commercialization of smallholder agriculture. The Plan for Accelerated and Sustainable Development to End Poverty (PASDEP), Ethiopia's strategic framework for 2005/06 – 2009/10, relies on a massive push to accelerate growth. This is to be achieved by efforts in two directions: commercialization of agriculture, based on supporting the intensification of marketable farm products (both for domestic and export markets, and by both small and large farmers); and promoting much more rapid non-farm private sector growth (MoFED, 2015).

2.1.3 Definition of commercialization

In most literature, a farm household is assumed to be commercialized if it is producing a significant amount of cash commodities, allocating a proportion of its resources to marketable commodities, or selling a considerable proportion of its agricultural outputs (MoARD 2010). However, the meaning of commercialization goes beyond supplying surplus products to markets (Pauw,2017). According to these authors, it has to consider both the input and output sides of production, and the decision-making behavior of farm households in production and marketing simultaneously. Moreover, commercialization is not restricted only to cash crops as traditional food crops are also frequently marketed to a considerable extent (Mebrahatom,2014). Commodities traditionally considered as food crops may increasingly be marketed during the transformation process as households specialize. The commonly accepted concept of commercialization is, therefore, that commercialized households are targeting markets in their production decisions, rather than being related simply to the amount of product they would likely sell due to surplus production (Tekalign,2014). In other words, production decisions of commercialized farmers are based on market signals and comparative advantages, whereas those of subsistence farmers are based on production feasibility and subsistence requirements, and selling only whatever surplus product is left after household consumption requirements are met.

According to Tadele and wudineh (2017), commercialization of subsistence agriculture takes many forms. They state that: “Commercialization can occur on the output side of production with increased marketed surplus, but it can also occur on the input side with increased use of purchased inputs. Commercialization is not restricted to just cash crops: The so called traditional food crops are frequently marketed to a considerable extent, and the so-called cash crops are retained, to a substantial extent, on the farm for home consumption, as, for instance, groundnuts in West Africa. Also, increased commercialization is not necessarily identical with expansion of the cash economy when there exist considerable inland transactions and payments with food commodities for land use or laborers. Finally, commercialization of agriculture is not identical with commercialization of the rural economy.”

The Millennium Development Project’s Hunger Task Force concluded in 2005 that “the world could meet the MDG of halving hunger by 2015”, and that “development of agriculture is critical to that goal” (ATA,2016). Rural areas are the home of the majority in Africa and small scale agriculture is the mainstay of the rural economy serving mainly as a source of food income (ATA,2016). The literature shows that “with the adoption of improved technologies and modern techniques, access to agricultural inputs and investment in infrastructure, rapid growth in agricultural incomes is achievable in Africa” (Tura,2016). Smallholder agriculture, which is the predominant source of livelihoods in Africa, has proven to be as at least as efficient as larger farms when farmers have received similar support services and inputs (seed, fertilizer, and credit) (Alemu,2015).

Many countries and international development agencies give due concern to intensification and commercialization of smallholder agriculture as a means of achieving poverty reduction; and thus they have reflected it in their official policies (ATA,2016).

In Ethiopia, there were many attempts to integrate the farmers into the market since the 1950s. In the 1950s the emphasis had been on improving productivity and reducing economic dependence on agriculture whereas in the 1960s, it shifted to agro-industrial economy and increment of foreign earnings (CSA,2016). In the 1970’s the focus shifted to small holder potential after inefficiencies were observed in mechanized farms. In the 1980’s the country adopted the socialist agricultural development strategy following the rise of the Derg regime to power. Since the coming to power of the current government in the 1990s, strong focus has been given to

smallholder farming and poverty reduction, and supporting agricultural intensification (Pauw,2017).

According to MoFED, the Ethiopian government has prioritized commercialization of farming as a policy agenda since 2015 and this priority is demonstrated by the central place this issue has gained in the second Poverty Reduction Strategy Paper (PRSP) (Habtewold,2017). The second Poverty Reduction Strategy Paper for Ethiopia (PRSP), known as the Plan for Accelerated and Sustainable Development to End Poverty (PASDEP), is established up on eight pillars; the second pillar intends to achieve growth and thereby improve people's livelihoods and reduce poverty (Kim,2016). The plan has set out two directions in order to achieve the aforementioned objectives: "commercialization of agriculture, based on supporting the intensification of marketable farm products (both for domestic and export markets, and by both small and large farmers); and promoting much more rapid non-farm private sector growth" (MoFED, 2015).

According to Tura (2016), the Ethiopian government has shown commitment to re-orient smallholder farmers from subsistence to market focused production while also strengthening the development of large-scale and export oriented farming ventures to seize the benefits of large-scale production systems. There are approximately 11.5 million smallholders in Ethiopia (Alemu,2015).

The Ministry of Agriculture and Rural Development (MoARD) has, under its 2010 master plan for enhanced market-oriented production, identified several crops, viz. teff, wheat, barley, lentil, chickpea, haricot beans, cotton, sesame, coffee and spices as priority crops (MoARD, 2010 cited in Samuel and ATA,2015). Accordingly, "the rural development strategy intends to contribute to the transformation of the productive rural sector from a primarily subsistence oriented to a more market-oriented sector, contributing to overall economic growth and poverty reduction" (Alemu,2015)

According to ATA (2016), the final intention of going commercial is not just making a shift from subsistence to market oriented farming but, by doing so, to achieve better welfare outcomes for the smallholders. To the minimum, welfare can be represented by increased consumption of basic and high valued food (livestock products), higher expenditure on education, healthcare, shoes and clothes and durable goods.

2.1.4 Teff production in Ethiopia

Ethiopia is commonly regarded as the only country in the world where teff is a significant crop. In Ethiopia it is particularly popular in the western provinces (Leykun and Jema,2014).

Teff is a resilient crop, and in areas where it can be cultivated most farmers grow it as a security crop. During the Derg regime, from 1974 to 1991, the cultivation of teff was discouraged, as it was considered to be of little value in terms of nutrition. Wheat, sorghum and maize were promoted as alternatives. However, this policy did not put an end to teff cultivation in Ethiopia. Whereas in 1960, approximately 40% of the country's total area used for cereal cultivation was devoted to teff, by 1980 this figure had grown to 50% (Leykun,2014). And despite various policy challenges, teff has remained important for most Ethiopians: it is often the only crop that survives and bears grain during difficult periods.

According to Pauw (2017) the Ethiopian germ plasm collection held 3892 teff accessions in 2005. He also refers to the existence of 2255 pure-line accessions. The IBC gene bank holds 4540 teff accessions, although some of these may be copies of the same accession. Since 1970, 32 varieties of teff have been released in Ethiopia. Eighteen of these have been released by Debre Zeit Agricultural Research Centre, with the rest coming from other federal research centers (two from Holetta Agricultural Research Centre and one from Melkassa Agricultural Research Centre) and regional research institutes.

2.2. Review of Empirical Evidences

2.2.1 Teff in the Ethiopian economy

Teff is particularly interesting in the context of smallholder commercialization and food security, since it has high value as both cash and a food crop. Many poorer farmers with suitable land grow it almost entirely for sale, using the proceeds to buy cheaper staples; although, as they become more prosperous, they may retain more for their own consumption. It is grown entirely by smallholders, and has been actively marketed for many generations. Until recently its market was almost wholly domestic, within Ethiopia (and formerly Eritrea): however, a promising niche export market is now developing in Europe and America, based on teff's increasing reputation as a "super-grain", being gluten-free and high in protein and calcium as well as micronutrients such

as iron and B vitamins. Farmers' preference for growing teff is due to production characteristics as well as consumer demand. The crop has a wide altitude range, and its resistance to diverse biotic and abiotic stresses makes it "low-risk" for cultivation (Osmani, 2014). It also stores well, since the very small size of the grain makes it resistant to post-harvest damage by insects. Among Ethiopia's 11.3 million small grain farmers, about 46% (5.2 million) grew teff in 2005/06 (CSA, 2016).

This makes teff the second most widely-grown annual crop after maize, which was cultivated by 6.8 million farmers. During the same year, teff was grown on over 2.24 million hectares which is a little over one fifth (21%) of the total land planted to grains. The average farmer cultivated teff on 0.43 hectare, on which he or she produced 4.2 quintals², implying an average yield of 9.8 quintals per hectare. The national production of teff has increased tremendously over the last twenty years, from 11.8 to 21.8 million quintals (a rise of nearly 85%). However, this encouraging performance must be interpreted in the context of high population growth and poor conditions in the base year. Per capita production grew by only 23% (1.9% per year) over the same period, and has never exceeded 30 kg.³ It is also a matter of concern that most (64%) of the growth in production is attributable to area expansion, while improved yields contributed only 12%.⁴ This indicates the enormous difficulty of achieving broad-based agricultural productivity growth, which is critical to lift the majority out of poverty. In Ethiopia, Cereal production and marketing are the means of livelihood for millions of smallholder households and it constitutes the single largest sub-sector in economy. Cereal accounts for roughly 60% of rural employment, 80% of total cultivated land, more than 40% of typical household's food expenditure, and more than 60% of total caloric intake. The contribution of cereals to national income is also large. According to available estimate, cereal production represents about 30% of gross domestic product (GDP). This calculation follows from the fact that agriculture is 48% of the nation's GDP (World Bank, 2013), and that cereals' contribution to agricultural GDP is 65% (Diao et al, 2012).

In the country, cereals are also the major staple food crops taking a significant share of area cultivated and volume of production obtained. Out of the total grain crop area, 79.69% (8.7 million hectares) was covered by cereals. Teff covered up 23.42% (about 2.7 million hectares) of grain crops. Cereals also contributed to 85.11% (about 137.1 million quintals) of the

total grain production. The contribution of teff was 18.57% (34.8 million quintals) of the total Meher cereals produced in the same order (CSA, 2016).

3. METHOD AND METHODOLOGY

3.1 DiscriptionOf The Study Area

The study was conducted in Bedele. It is one of the woreda in the [Oromia Region](#) State of Ethiopia. Bedele is one of the south western BunnoBedele zones of oromia regional state. Bedele is bordered on the south by Gechi and on the south west by Chora, on the north by southern BenishangulGumuz region, on the North West by Didessa river which separates it from the east Welega zone. Located in the Buno Bedele Zone of the oromia region about 492 km from Addis Ababa the capital city of the countries, 602 km from Adama, 90 km from nekemte, 120 km from Mettu. It characterized by warm climate with an average annual rainfall of about 790-866mm, annual mean temperature of 18.4 degrees centigrade with mean minimum and maximum temperature of 17.4 and 23.3 degrees centigrade respectively. The latitudes and longitudes of 827N and 36 21 E respectively. Its altitude ranges from 1600-2100 meters above sea level.

According to 2020 national census reported a total population for Bedele is 118,157 of total population, 58,510 were men and 59,647 were women. The majority of the inhabitants practiced [Ethiopian Orthodox Christianity](#), with 52.98% reporting that as their religion, while 22.04% of the populations were Muslim, and 24.98% of the populations were protestant. The majority of the society engaged and their income is mainly depends on agriculture. The crop grow in the area is coffee, wheat, teff, sorghum, maize, chick pea, barely mug bean and from vegetable horticultural crop cabbage, tomato onion, garlic, carrot and orange, lemon, banana, are commonly grown in the area.

3.2. Sampling Techniques and Sample Size

The target population for this study was all households of Bedele district. The simple random sampling technique has applied for gathering necessary information for the study. The reason for selecting simple random sampling method will be the fact that the Bedele district has above 80% homogeneous population. The woreda have 15 kebeles. From these kebeles, I want to select three

kebeles. These were namely, DabanaDaru,Ourgesa and Tabba chabali, because some household in these kebele are commercialization of teff as compared to the other kebeles.

Therefore, Dabana Daru has total household of 657, Ourgesa has total household of 481 and Tabba chabali has total household of 454. In order to take a total sample size relevant to target study I apply Yamane formula (1967) of $n = \frac{N}{1 + N(e)^2}$. In addition to this, we are going to use proportionate sampling that was enable us in taking samples from each sub strata' which are to be selected from the target population.

$$N = 657 + 481 + 454$$

where, N = the total number of households

$$N = 1592$$

e = the margin of error

$$e = 10\% \text{ or } 0.1$$

n = the total sample size of households

$$n = \frac{N}{1 + N(e)^2}$$

$$n = \frac{1592}{1 + 1592(0.1)^2}$$

$$n = 94$$

We has applye proportionate sampling to obtain how many respondents are selected from each stratum by using formulas: $nh = nh(Nh)/N$

Where: nh = Sample size for stratum h

Nh= is the numbers of households for stratum h

N= is the total numbers of households and n is the total sample size.

$$nh1 = n(Nh1)/N$$

$$nh2 = n(Nh2)/N$$

$$nh3 = n(Nh3)/N$$

$$= 94(657)/1592$$

$$= 94(481)/1592$$

$$= 94(454)/1592$$

$$nh1 = 39$$

$$nh2 = 28$$

$$nh3 = 27$$

So that, $nh_1=39$, $nh_2=28$, and $nh_3 = 27$ are the sample size of the strata to the kebele DabanaDaru, Ourgesa, and TabbaChabali respectively.

3.3. Types and Source of Data

Both qualitative and quantitative data type were collected from different source. The sources of data are both primary and secondary sources. The major data collection methods were discussions with individuals, focus groups, rapid market appraisal, observation and formal survey.

3.4 Methods of Data Collection

To collect the quantitative and qualitative data, this study was using the following main instruments namely questionnaire, interview and organizational document from quantitative data collection techniques and key informant interview from qualitative data collection techniques.

Primary data collection method: firstly, the whole situations of the marketing system from the producer farmer up to the end consumer assessed through rapid market appraisal. Informal survey was employ on rapid market appraisal (RMA) technique using check list. Secondly, in order to collect primary data a semi-structural questionnaire prepared. The questionnaires were pre-tested and revised based on the feedback. Finally, focus groups (discussion and interviews) were also hold with input supplier, farmer, traders, processor, consumers and agricultural expertise from both governmental and non-governmental organization. Moreover, checklist prepared to guide focus group discussions. Primary data on teff producer, the existing market situation and linkages, volume of trade, market institutions and facilities, marketing problems, credit facilities, household characteristics (sex, education level, farming experience etc.), role of marketing agents, selling and buying strategies, marketing facilities, and other socio-economic variables were collected by using semi-structure questionnaire.

Secondary data collection method: Secondary data collect from kebele and district level agriculture and rural development offices, district micro finance office, Ethiopian revenue and customs authority, central statistical agency.

3.5. Methods of Data Analysis

In this study two types of data analysis, viz. descriptive statistics and econometric analysis were used for analyzing the data collected to meet the objectives of the study.

3.5.1. Descriptive analysis

By employing descriptive statistics I can compare differences among categories of sample units with respect to the desired characteristics. In this study, descriptive statistics such as mean, percentages and frequency of occurrence were used along the econometric model, to analyze the collected secondary and primary data. A multiple linear regression model was used to identify the determinants of level of teff commercialization among the farmers (producer). The most common approach used in measuring the level of teff commercialization at the household level has been using the proportion of sales from the total value of agricultural production (von Braun 1994). This is actually the revealed marketing decision of a household, particularly for commodities that are potentially used for sale and home consumption (Randolph 1992).

3.5.2. Econometric model

Multiple Linear regression models are used to model relationships between a dependent variable and a set of independent variables.

Regression Functions

The equation of regressions on this study is generally built around two sets of variables, namely dependent variable (teff commercialization) and independent variables (Distance to nearest market place (DISNMK), Level of teff Commercialization (LETC), Sex of the Household Head (SHH), Family size (FAMZ), Access to credit (ACCRD), Extension contact (EXCON), Farm experiences (FMEXP), Age of house hold (AGH) and Intensity of fertilizer (AVFZR). The basic objective of using regression equation on this study is to make the study more effective at describing, understanding and predicting the stated variables.

$$Y_i = \beta_0 + \beta_1 \text{DISNMK} + \beta_2 \text{SHH} + \beta_3 \text{FAMZ} + \beta_4 \text{ACCRD} + \beta_5 \text{AVFZR} + \beta_6 \text{EXCON} + \beta_7 \text{FMEXP} + \beta_8 \text{AGH} + \beta_9 \text{AVFZR}$$

Where:

Y_i = is the response or dependent variable- amount of teff.

DISNMK = Distance to nearest market place,

SHH= Sex of the Household Head,

FAMZ = Family size,

ACCRD= Access to credit,

AVFZR= Intensity of fertilizer use,

EXCON= Extension contact,

FMEXP= Farm experience.

AGH=Age of house hold.

Coefficient of Multiple Determinations

The multiple coefficient of determination represents the percentage of variability in Y that is explained by the estimated regression equation. We have $0 < r^2 < 1$ as in the case of simple regression. The positive square root of r^2 is the multiple correlation coefficients between Y and the set of control variables.

The dependent variable of the model

It is continuous dependent variable and measured in percentage. It represents the actual degree of teff marketed by farm households.

Independent (explanatory) variables

Distance to nearest market place (DISNMK): Distance to the market is a continuous variable measured in kilometers from the household residence to the market centers. The closer the residence of the household to the rural market center that reduces cost of transportation so the more was volume of teff marketed. The assumption here is that the closer a household is to the

market, the more the household will increase volume of teff marketed. Therefore, this variable was expected to have an inverse relationship with the teff commercialization.

Sex of the Household Head (SHH): this is a dummy variable, which takes a value of one if the household head is male and zero, if female. Literature indicate that female-headed households have less access to improved technologies, land, and extension services as compared to male headed households. If the household head is male, he has enough time compared to female headed to get more information about commercialization of teff. This variable is expected to have indeterminate effect on teff commercialization.

Family size (FAMZ): this variable is a continuous variable and refers to the total number of family members (adult equivalent) in the household. The size of economically active family members within a given farming household affects the crop production activities positively. Thus, in this respect family size is expected to have positive impact on commercialization. But larger family size requires larger amounts for consumption. Therefore, family size can affect teff commercialization either positively or negatively.

Access to credit (ACCRD): This is a dummy variable, which takes a value of one if the amount of credit received and utilized by farmers from different organizations like micro finance institution, cooperatives and other institutions operating in the area. According to Demeke (20014), credit makes traditional agriculture more productive through the purchase of farm equipment and other agricultural inputs, and technological developments and zero otherwise Credit can also be used as an instrument for market stability. Rural farmers can build their bargaining power by establishing storage facilities and providing transport system acquired through credit. Therefore, this variable is expected to have a positive relation to the level of teff commercialization.

Age (AG): this variable is treated as a continuous variable and Study findings support consideration of age of farmer as a factor that may affect their teff commercialization.

Extension contact (EXCON): this variable is treated as a continuous variable and measured by number of visits made by the extension agents to sample households. Extension visits were help to reinforce the message and enhance the accuracy of implementation of the technology packages (Demeke,2014). More frequent DA visits, using different extension teaching methods

like attending demonstrations, field day, and participation on on-farm trials can help the farmers to adopt a new technology. The finding of Tura (2016) identified that extension visit was negatively related to pepper market entry decision and marketed pepper volume. However, all farmers may not have equal access to extension services. If the farmers get better extension services, they are expected to adopt improved teff production technologies than others. Therefore contact with extension agents were positive direct relation with the level of commercialization.

Farming experiences (FMEXP): it is a continuous variable measured in number of years stayed in teff production and marketing. Household with better experience in teff production and marketing was expected to produce more amounts of teff and supply to the market than those with only less experience and, as a result, it is expected to supply more amounts to market. Therefore, experience in teff production was expected to have a positive relation with the level of commercialization.

Intensity of fertilizer (AVFZR): it is a continuous variable and measured in Kilogram. It refers the availability of fertilizer to teff producers. According to the finding of Tekalign(2016) intensity of fertilizer is not included his finding but If fertilizer is available in the right amount and time it was increase teff production. So directly the volume of marketed was increased. Hence, this variable was expected positive relation with teff commercialization.

Table 1: hypothesis the relation between independent variables and dependent variable

Variable	Type	measurement	Expectation
Distance to nearest market place	Continuous	Km	-ve
Sex of the Household Head	Dummy 0=F,1=M		-ve/+ve
Family size	Continuous	Number/household	+ve -ve
Access to credit	Dummy 0=own,1=non own		+ve
Intensity of fertilizer	Continuous	Qunta	+ve
Extension contact	Continuous	Number of visit	+ve
Farming experiences	Continuous	Number of years	-ve

Age of house hold	Continuous	Number of year	-ve
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4. BUDGET PLANNING

4.1 Budget setting

Budget plan shows the budget estimated for the accomplishment of the research Proposal.

Nº	Item description	Unit	Unit price	Amount of required	Total price
1.	Paper	Packet	90	1	90
2.	Pen	Number	15	15	225
3.	Memory card	Number	150	1	150
4.	Printing	Page	3.50	32	112
5.	Photo copy	Page	1	100	100
6.	Fixing	Number	6	2	12
7.	Telephone			Three months	100
8.	Internet				50
9.	Transport				200
10.	Sum/total				1039

5. WORKING PLAN

5.1 Time schedule

The research proposal will be conducted during 2022 within four months, starting from Mar up to Jun.

Activities	2022			
	Mar	Apr	May	Jun
Title selecting	√			
Proposal writing	√			
Data collection		√		
Data entry		√		
Data analysis		√		
1 st draft project writing			√	
Final draft project writing				
Project presentation				

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7.APPENDEXES.

MEU Bedele campus College of Natural Resource And Agricultural Economics Department Of Agribusiness and value chain management

Questions prepared for producers

I. General Information

1. Name of Respondent: _____
2. Zone: _____ Woreda _____ Kebele _____
3. Age of the respondent: _____ years
4. Sex of the respondent (✓): 1 ☐ Male ☐ Female
5. Education level of the respondent in years of schooling? _____
6. Marital status (✓): 1. [] Married 2. [] Unmarried 3. [] Divorce 4. [] Widowed
7. What is your major means of income generation? -----

8. For whom you sell?

A, to relative person B, to the hotels c, to the consumer

d, Household and Resource Data.

1. Family size: [] Male [] Female [] Total
Below 15 years [] Male [] Female []
Total
Between 15 and 64 years [] Male [] Female
[] Total

Above 65 years[_____] Male [_____] Female [_____]
Total

2. Do you own arable land? (✓) 1. [] Yes 2. [] No
3. If yes, how much in hectare? _____
4. Total cultivated land size:.....timad..... ha. (Note: 1 ha = 8 timad or 1 timad = 0.125 ha)
5. Total irrigable area:.....timad..... ha.

Supply of fertilizer

1. Have you ever used fertilizer for the production of teff? (✓) 1. [] Yes 2. [] No
2. If your answer for Q.1 is No, what was the main reason behind?.....
3. If your answer for Q.1 is yes, how much did you used in the teff production process last year?
_____ quintal.
4. Do you always get fertilizer at the right time? (✓) 1. [] Yes 2. [] No
5. If your answer for Q.4 is No, what are the reasons? (✓) 1. [] Unavailability 2. []
Far distance 3. [] Others (specify)

Farming experience

1. How long have you practiced production of teff products? _____ Years

Access to Credit

- 1'. Do you have an access to credit? (✓) 1. [] Yes 2. [] No
2. If your answer for Q.1 is Yes, how much do you get? _____ birr
3. If your answer for Q.1 is No. what are reasons?
1. [] Limited supply of credit 2. [] Limited access to transport
3. [] Huge bureaucracy 4. [] Others (specify)

Extension contact

1. Have you ever get extension contact on Teff production practices? (✓)

1. ☐ Yes 2. ☐ No

2. If your answer for Q.1 is No, why? (✓) (Multiple responses is possible)

1. ☐ No service provider nearby 2. ☐ Availability of contact farmers

3. ☐ Do not have time to get the service 4. ☐ Others (specify)

3. If your answer for Q.1 is Yes, How frequent were you visited by extension agents last year?
_____ visits

4. Who provides the extension contact? (✓) (Multiple response is possible)

1. ☐ Development agents 2. ☐ NGOs (specify)

3. ☐ Research centers (specify) 4. ☐ Neighbors and friends

5. ☐ Others (specify)

Distance to nearest market place and road

1. How long distance from nearest market place? _____ Km

2. Distance from the main road? _____ km

Quantity of teff produced and marketed

1. Have you engaged in the production of teff last year? 1, yes 2, no

2. If yes, how much? _____ quintal.

3. How much quintal of the produced amount is sold in the market?
_____ quintal.

4. If no, why? _____

What were the main challenges in teff marketing?

